



Data Converters

Introduction

Summer Term 2026

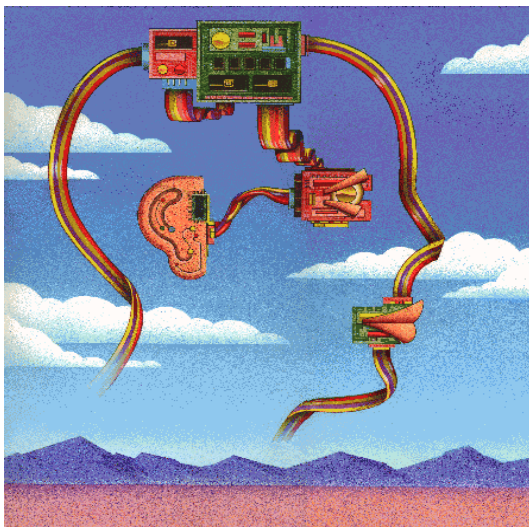
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Department of Microsystems Engineering – IMTEK
University of Freiburg



Our Real World is Analog!



We typically want to capture and process analog signals that are continuous in both time and amplitude.

Scientific American
1999

26-04-22

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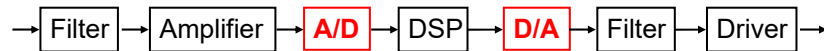
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Motivation

However, we want to perform digital signal processing.
It is digital circuit design that benefits from technology scaling
in terms of power and area.



A/D and D/A converters are fundamental mixed-signal circuit blocks
bridging our analog world and the digital domain of signal processing.

They will always be needed and
never be replaced by all-digital circuit blocks.



Intention of the Course

The course covers

- the fundamentals of data conversion, i.e., filtering, sampling, and quantization for A/D conversion and digital-to-analog conversion, analog hold, and reconstruction for D/A conversion
- the static and spectral metrics and nonidealities of A/D and D/A converters, e.g., gain/offset error, integral/differential nonlinearity, dynamic range, signal-to-noise(-and-distortion) ratio, etc.
- an overview and discussion of state-of-the-art Nyquist D/A converters
 - an overview and discussion of state-of-the-art Nyquist and oversampled A/D converters.



General Information

Lecture (3 ECTS): Thursday 12:15pm – 1:45pm, room 101-009/13

Exercise (3 ECTS): Thursday 2:15pm – 3:45pm, room 101-009/13

2h written exam: in the examination period, date to be announced

All Information concerning this course
<http://www.imtek.de/mikroelektronik>

→ Teaching → Summer Term 2026 → Data Converters

→ Lehre → Sommersemester 2026 → Data Converters

Recordings of the lectures / materials / etc. → ILIAS

No recordings of the exercises!

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Course Plan

Week	Date	Lecture - Recordings in ILIAS	Exercise – No Recordings
1	26/04/23	Chap I (90 min)	Chap I (20 min) / Chap II (45 min) / Chap III (25 min)
2	26/04/30	Chap III (45 min) / Chap IV (45 min)	Discussion PE I
3	26/05/07	Chap IV (55 min) / Chap V (35 min)	Chap V (90 min)
4	26/05/14	Ascension Day (no lecture)	Ascension Day (no exercise)
5	26/05/21	Chap V (90 min)	Discussion PE II
6	26/05/28	Pentecost Break (no lecture)	Pentecost Break (no exercise)
7	26/06/04	Chorpus Christi (no lecture)	Chorpus Christi (no exercise)
8	26/06/11	Chap V (finish)	Discussion PE III
9	26/06/18	Chap VI (90 min)	Discussion PE IV
10	26/06/25	Chap VI (30 min) / Chap VII (60 min)	Discussion PE V
11	26/07/02	Chap VII (90 min)	Discussion PE VI
12	26/07/09	Chap VIII (90 min)	Discussion PE VII
13	26/07/16		Discussion PE VIII
14	26/07/23		

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Acknowledgement

Concepts & slides are partially based on the lecture

VLSI Data Conversion Circuits

held by

Prof. B. Murmann, PhD, University of Hawai'i

during his sabbatical at the University of Freiburg in 2009

